

Proximal Gradient Algorithm

$$\min: f(x) = \underset{\substack{\uparrow \\ \text{smoother}}}{g(x)} + \underset{\substack{\uparrow \\ \text{easy to evaluate}}}{h(x)}$$

prox

$$\underline{x_{t+1} = \text{prox}_{\eta h}(x_t - \eta \nabla g(x_t))}$$

// if $\boxed{h(x) = \mathcal{I}_{\mathcal{X}}(x)}$ then
prox-gradient becomes
projected gradient.

Properties of Prox

Recall that for projection: $\|y_1 - y_2\|_2 \geq \|\text{Proj}_X(y_1) - \text{Proj}_X(y_2)\|_2$

Similarly: $\|\text{prox}_h(x) - \text{prox}_h(y)\|_2 \leq \|x - y\|_2$. $\leftarrow$ follows from $\hookrightarrow$ using C-S.

$$\langle \text{prox}_h(x) - \text{prox}_h(y), x - y \rangle \geq \|\text{prox}_h(x) - \text{prox}_h(y)\|_2^2$$

let $u = \text{prox}_h(x)$ $v = \text{prox}_h(y)$

$$u = \text{prox}_h(x) = \underset{u}{\text{argmin}}: h(u) + \frac{1}{2} \|x - u\|_2^2$$
$$0 \in \partial h(u) + (u - x)$$

$$\Leftrightarrow \boxed{(x - u) \in \partial h(u)} \quad \text{also} \quad \boxed{(y - v) \in \partial h(v)}$$

h convex $\Rightarrow \partial h$ monotone

$$\Rightarrow \langle (x - u) - (y - v), u - v \rangle \geq 0$$

$$\Rightarrow \langle x - y, u - v \rangle \geq \|u - v\|_2^2$$

$$\Rightarrow \langle x - y, \text{prox}_h(x) - \text{prox}_h(y) \rangle \geq \|\text{prox}_h(x) - \text{prox}_h(y)\|_2^2.$$

Proximal Gradient - Gradient mapping

For smooth functions, key inequality:

For GD of smooth g - $\boxed{x_{t+1} = x_t - \eta \nabla g(x_t)}$

$$g(x_{t+1}) = g(x_t - \frac{1}{\beta} \nabla g(x_t)) \leq g(x_t) - \frac{1}{2\beta} \|\nabla g(x_t)\|_2^2 \quad (\text{from quadratic upper bound})$$

$$\text{Now: } x_{t+1} = \text{prox}_{\eta h}(x_t - \eta \nabla g(x_t))$$

$$\boxed{\min: f(x) = g(x) + h(x)}$$

The gradient mapping: Def'n: $G_\eta(x) = \frac{1}{\eta} (x - \text{prox}_{\eta h}(x - \eta \nabla g(x)))$

This gives: $\boxed{x_{t+1} = x_t - \eta G_\eta(x_t)}$

Note: $G_\eta(x_t)$ in general is not $\in \partial f(x_t)$

Proximal Gradient

Recall:

$$G_\eta(x) = \frac{1}{\eta} (x - \text{prox}_{\eta h}(x - \eta \nabla g(x)))$$

Optimal solutions are the only fixed points of the prox grad update

$$x_{t+1} = x_t - \eta G_\eta(x_t)$$

equivalently: $G_\eta(x) = 0 \iff x$ minimizes $f(x) = g(x) + h(x)$

To see this:

$$G_\eta(\hat{x}) = 0 \iff \hat{x} = \text{prox}_{\eta h}(\hat{x} - \eta \nabla g(\hat{x}))$$

$$\iff (\hat{x} - \eta \nabla g(\hat{x})) - \hat{x} \in \eta \partial h(\hat{x})$$

$u = \text{prox}_{\eta h}(x)$: $0 \in \partial h(u) + \frac{1}{\eta}(u - x)$
 $(x - u) \in \eta \partial h(u)$

$$\iff -\eta \nabla g(\hat{x}) \in \eta \partial h(\hat{x})$$

$$\iff \boxed{0 \in \nabla g(\hat{x}) + \partial h(\hat{x})}$$

$$\min: g(x) + h(x) \quad \hat{x} \text{ opt sol'n} \iff 0 \in \nabla g(\hat{x}) + \partial h(\hat{x})$$

Proximal Gradient - key lemma

Need analog: $g(x - \frac{1}{\beta} \nabla g(x)) \leq g(x) - \frac{1}{2\beta} \|\nabla g(x)\|_2^2$.

Key lemma:

$f = g + h$, g is β -smooth, α -strongly convex, (α may be zero)

$$\eta \leq \frac{1}{\beta}$$

Lemma: $f(x - \eta G(x)) \leq f(z) + \langle G_\eta(x), x - z \rangle - \frac{1}{2} \|G_\eta(x)\|_2^2$
 $- \frac{\alpha}{2} \|x - z\|_2^2$.

Consequence: analog of our basic inequality

Use Lemma with: $z = x_t$, $x = x_t$

$$f(x_{t+1}) = f(x_t - \eta G(x_t)) \leq f(x_t) - \frac{1}{2} \|G_\eta(x_t)\|_2^2$$

compare to inequality at top.

Proximal Gradient

$$\min: f(x) = \underset{\substack{\uparrow \\ \beta\text{-smooth} \\ \alpha\text{-strongly convex}}}{g(x)} + \underset{\substack{\uparrow \\ \text{convex, easy-to-evaluate} \\ \text{prox operator}}}{h(x)}$$

$$x_{t+1} = \text{prox}_{\eta h}(x_t - \eta \nabla g(x_t)), \quad \eta = 1/\beta$$

$$G_{\eta}(x) = \text{gradient mapping} = \frac{1}{\eta}(x - \text{prox}_{\eta h}(x - \eta \nabla g(x)))$$

$$\underline{x_{t+1} = x_t - \eta G_{\eta}(x_t)}$$

Key Lemma: $f(x - \eta G_{\eta}(x)) \leq f(z) + \langle G_{\eta}(x), x - z \rangle - \frac{\eta}{2} \|G_{\eta}(x)\|_2^2 - \frac{\alpha}{2} \|x - z\|_2^2$

Proximal Gradient - rate of convergence (1/4)

proof of key lemma (1/2)

$$f(x - \eta G_\eta(x)) = \underbrace{g(x - \eta G_\eta(x))}_{(a)} + \underbrace{h(x - \eta G_\eta(x))}_{(b)}$$

$$\begin{aligned} (a) &\leq g(x) - \eta \langle \nabla g(x), G_\eta(x) \rangle + \frac{\eta}{2} \|G_\eta(x)\|_2^2 \quad (\text{smoothness of } g) \\ &\leq g(x) - \langle \nabla g(x), x - x \rangle - \frac{\alpha}{2} \|x - x\|_2^2 - \eta \langle \nabla g(x), G_\eta(x) \rangle + \frac{\eta}{2} \|G_\eta(x)\|_2^2 \quad (s.l.) \end{aligned}$$

(b) To upper bound $h(x - \eta G_\eta(x))$, we need element of $\partial h(x - \eta G_\eta(x))$

Recall: $u = \text{prox}_{\eta h}(x) \Leftrightarrow x - u \in \eta \partial h(u)$

$$\underline{x - \eta G_\eta(x)} = x - \eta \left(\frac{1}{\eta} (x - \text{prox}_{\eta h}(x - \eta \nabla g(x))) \right) = + \underline{\text{prox}_{\eta h}(x - \eta \nabla g(x))}$$

$$\Rightarrow x - \eta G_\eta(x) = \text{prox}_{\eta h}(x - \eta \nabla g(x))$$

$$\Rightarrow \boxed{G_\eta(x) - \nabla g(x) \in \partial h(x - \eta G_\eta(x))}$$

Proximal Gradient - rate of convergence (2/4)

proof of key lemma (2/2)

We just saw: $G_\eta(x) - \nabla g(x) \in \partial h(x - \eta G_\eta(x))$

$$\textcircled{b} \quad h(\underline{x - \eta G_\eta(x)}) \leq h(\underline{z}) - \langle G_\eta(x) - \nabla g(x), z - (x - \eta G_\eta(x)) \rangle$$

Combine $\textcircled{a} + \textcircled{b}$

We find:

$$\begin{aligned} f(x - \eta G_\eta(x)) &\leq \textcircled{a} + \textcircled{b} \\ &= f(z) + \langle G_\eta(x), x - z \rangle - \frac{\eta}{2} \|G_\eta(x)\|_2^2 - \frac{\alpha}{2} \|x - z\|_V^2 \end{aligned}$$

Proximal Gradient - rate of convergence (3/4)

Smooth but not strongly convex : $\boxed{f(x) = g(x) + h(x),} \quad \alpha = 0.$

Lemma with $z = x^*, x = x_t$:

$$f(x_{t+1}) = f(x_t - \eta G_\eta(x_t)) \leq f(x^*) + \langle G_\eta(x_t), x_t - x^* \rangle - \frac{\eta}{2} \|G_\eta(x_t)\|_2^2$$

$$\left(\text{compare : } f(x_{t+1}) - f(x^*) \leq \langle \nabla f(x_t), x_t - x^* \rangle - \frac{\eta}{2} \|\nabla f(x_t)\|_2^2 \right)$$

$$\rightarrow f(x_{t+1}) - f(x^*) \leq \langle G_\eta(x_t), x_t - x^* \rangle - \frac{\eta}{2} \|G_\eta(x_t)\|_2^2$$

$$\leq \frac{1}{2\eta} \left(\|x_t - x^*\|_2^2 - \|x_t - x^* - \eta G_\eta(x_t)\|_2^2 \right)$$

$$= \frac{1}{2\eta} \left(\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 \right)$$

Telescoping.

$$\begin{aligned} f(x_T) - f(x^*) &\leq \frac{1}{T} \sum_{t=1}^T (f(x_t) - f(x^*)) \leq \frac{1}{2\eta T} \left(\|x_1 - x^*\|_2^2 - \|x_T - x^*\|_2^2 \right) \\ &\leq \frac{1}{2\eta T} \cdot \|x_1 - x^*\|_2^2 \quad O(1/T) \end{aligned}$$

Proximal Gradient - rate of convergence (4/4)

smooth and strongly convex

$$\begin{aligned} 0 \leq f(x_{t+1}) - f(x^*) &\leq \langle G_\eta(x_t), x_t - x^* \rangle - \frac{\eta}{2} \|G_\eta(x_t)\|_2^2 - \frac{\alpha}{2} \|x_t - x^*\|_2^2 \\ &= \frac{1}{2\eta} \left(\|x_t - x^*\|_2^2 - \eta\alpha \|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 \right) \\ &= \frac{1}{2\eta} \left((1 - \eta\alpha) \|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 \right) \end{aligned}$$

$$\|x_{t+1} - x^*\|_2^2 \leq (1 - \eta\alpha) \|x_t - x^*\|_2^2$$

$$\eta = 1/\beta$$

$$= (1 - \alpha/\beta) \|x_t - x^*\|_2^2$$

$$= (1 - \alpha/\beta)^t \cdot \|x_0 - x^*\|_2^2$$

$$\boxed{\kappa = \beta/\alpha}$$

Linear convergence